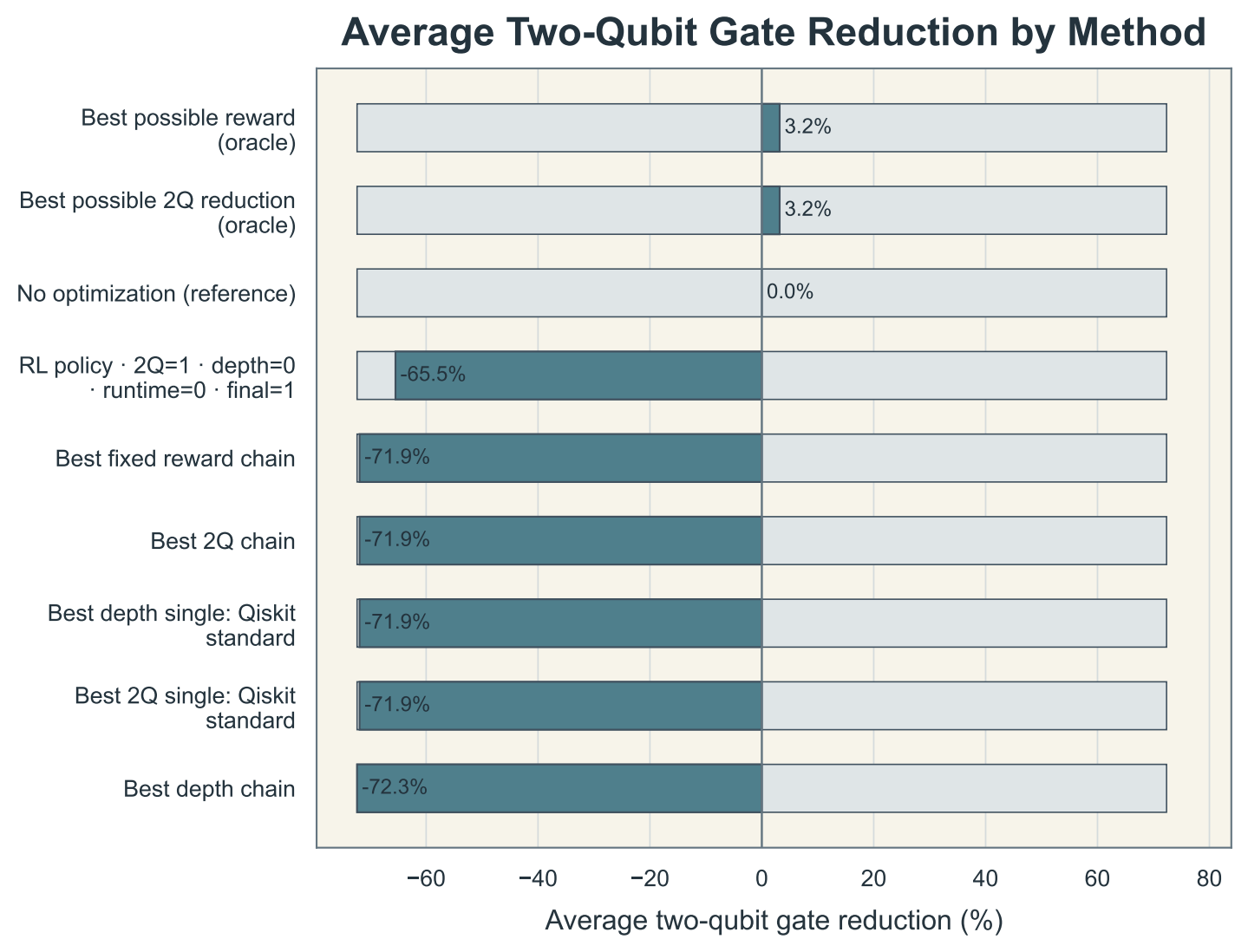
Average Two-Qubit Gate Reduction by Method
Best possible reward (oracle)
Best possible 2Q reduction (oracle)
No optimization (reference)
RL policy · 2Q=1 · depth=0 · runtime=0 · final=1
Best fixed reward chain
Best 2Q chain
Best depth single: Qiskit standard
Best 2Q single: Qiskit standard
Best depth chain
3.2%
3.2%
0.0%
-65.5%
-71.9%
-71.9%
-71.9%
-71.9%
-72.3%
−60
−40
−20
0
20
40
60
80
Average two-qubit gate reduction (%)